Project report on

2-bit Flash ADC

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Chapter 1: Introduction

Flash ADC (Analog to Digital Converter) is one of the fastest types of ADC architectures due to its parallel operation. It converts analog input signals into digital output without using iterative methods, making it suitable for high-speed applications.

In this project, a 2-bit Flash ADC is designed and simulated using the Microwind/DSCH tool. Compared to higher resolution ADCs, a 2-bit ADC is simple and requires fewer components, making it ideal for understanding the basic working of flash architecture.

A 2-bit Flash ADC consists of:

- 3 comparators
- A resistor divider network
- A priority encoder

The input analog signal is simultaneously compared with multiple reference voltages, and the outputs are converted into binary form using encoder logic.

This report focuses on the circuit design, working principle, layout implementation, and simulation results of the 2-bit Flash ADC.

Chapter 2: Circuit and its working

2.1 Block Diagram of Flash ADC

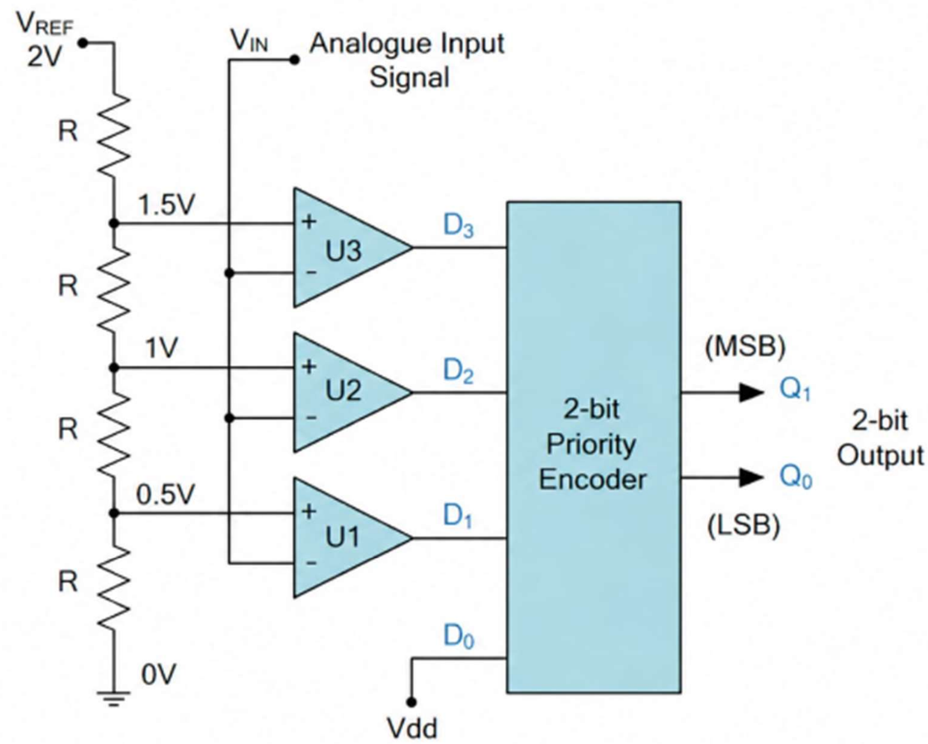

Fig. 1: 2-bit Flash ADC Circuit

2.2 Voltage Reference Circuit

A resistor divider network is used to generate reference voltages.

For a 2-bit ADC:

- Total levels = 4
- Reference voltages required = 3

These are:

- $V_1 = V_{ref}/4$
- $V_2 = V_{ref}/2$
- $V_3 = 3V_{ref}/4$

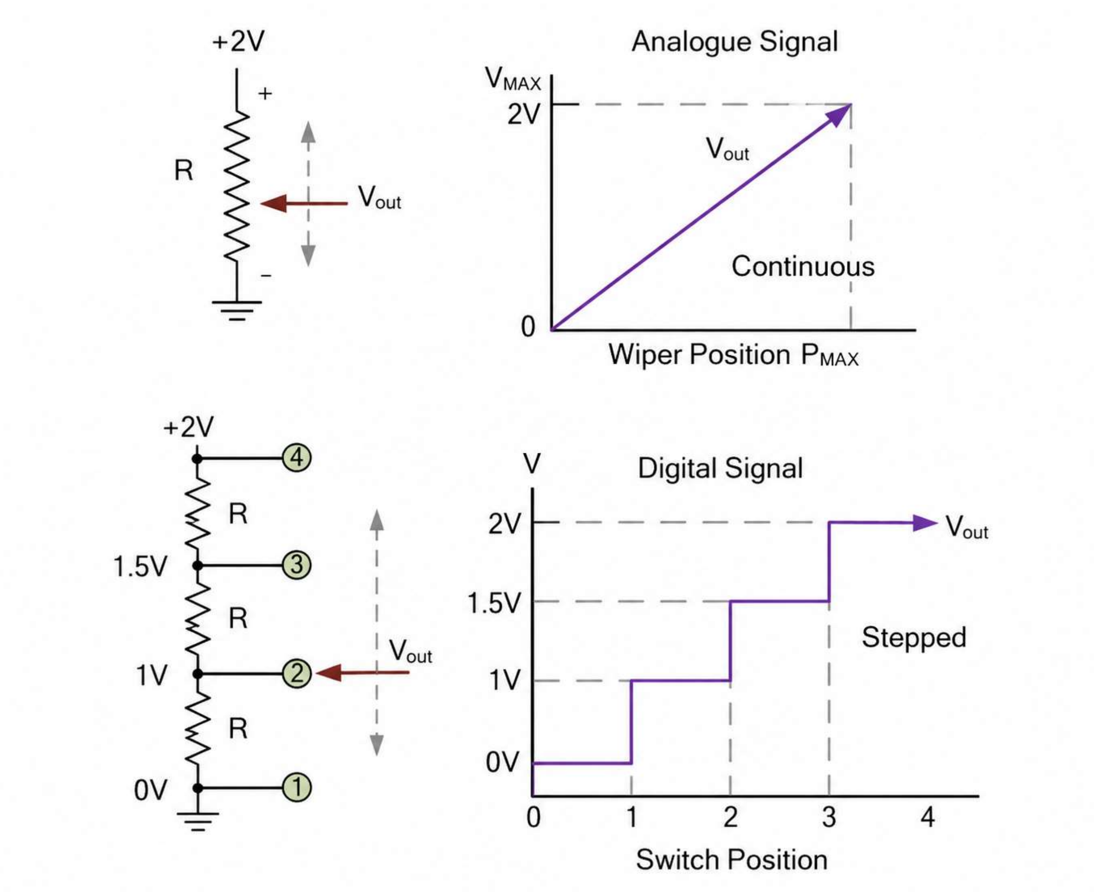

Fig. 2: Voltage Divider Circuit

2.3 Comparator Circuit

Comparators compare input voltage with reference voltages.

- If $V_{in} > V_{ref} \rightarrow$ Output HIGH
- Else → Output LOW

Each comparator produces one bit of thermometer code.

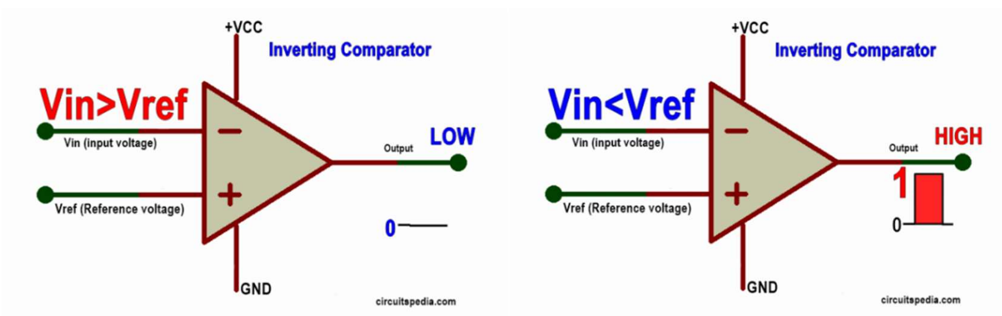

Fig. 3: 2-bit C

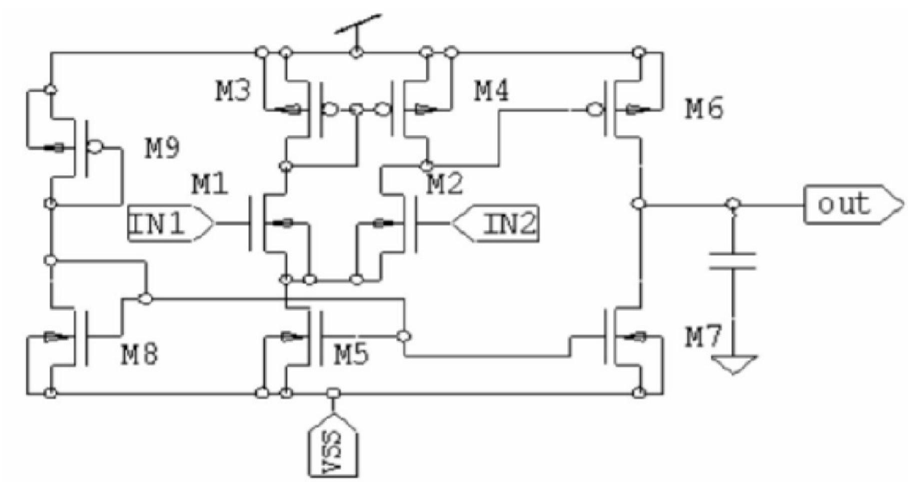

Fig. 4: Comparator using PMOS & NMOS System

2.4 Thermometer Code

The comparator outputs form thermometer code:

Vin Range	Output (C1 C2 C3)
Vin < V1	000
V1 < Vin < V2	100
V2 < Vin < V3	110
Vin > V3	111

2.5 Priority Encoder

Thermometer code is converted into binary output:

Thermometer	Binary Output
000	00
100	01
110	10
111	11

Logic Used:

$$\text{Output 1 (MSB)} = C_1 + C_2$$
$$\text{Output 2 (LSB)} = C_1 + \overline{C_2} \cdot C_3$$

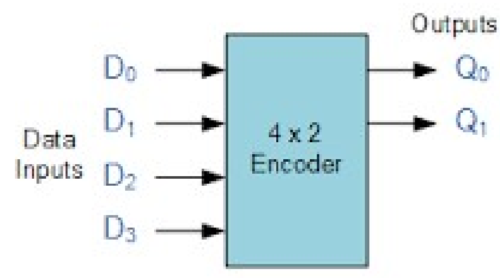

Fig. 5: Priority Encoder

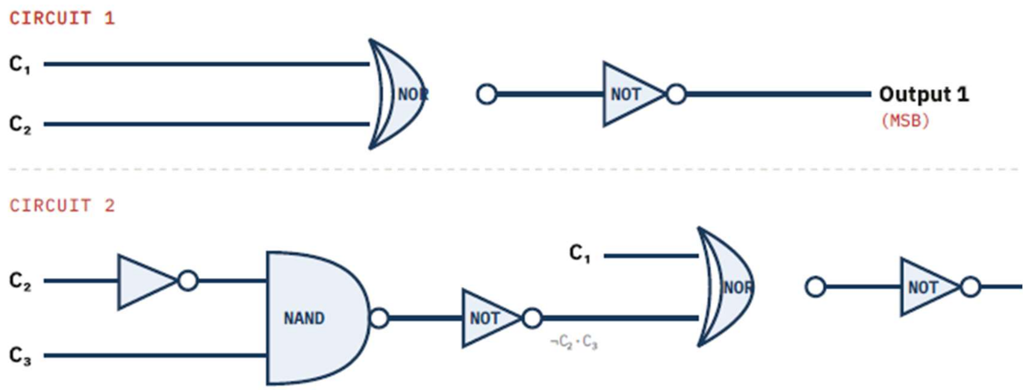

Fig. 6: Priority Encoder using gates

Gate	Inputs	Output Expression	Circuit	Role
NOR_1	C_1, C_2	$\neg(C_1 + C_2)$	Circuit 1	First stage of OR
NOT_1	$\neg(C_1+C_2)$	$C_1 + C_2$	Circuit 1	Produces MSB output
NOT_2	C_2	$\neg C_2$	Circuit 2	Invert C_2
NAND	$\neg C_2$, C_3	$\neg(\neg C_2 \cdot C_3)$	Circuit 2	First stage of AND
NOT_3	$\neg(\neg C_2 \cdot C_3)$	$\neg C_2 \cdot C_3$	Circuit 2	AND of $\neg C_2$ and C_3
NOR_2	C_1, $\neg C_2 \cdot C_3$	$\neg(C_1 + \neg C_2 \cdot C_3)$	Circuit 2	First stage of OR
NOT_4	$\neg(C_1 + \neg C_2 \cdot C_3)$	$C_1 + \neg C_2 \cdot C_3$	Circuit 2	Produces LSB output

2.6 Working of ADC

1. Input voltage is applied to all comparators
2. Comparators compare input with reference voltages
3. Outputs form thermometer code
4. Encoder converts it into binary output

Chapter 3: Layout and Results

3.1 Priority Encoder Layout

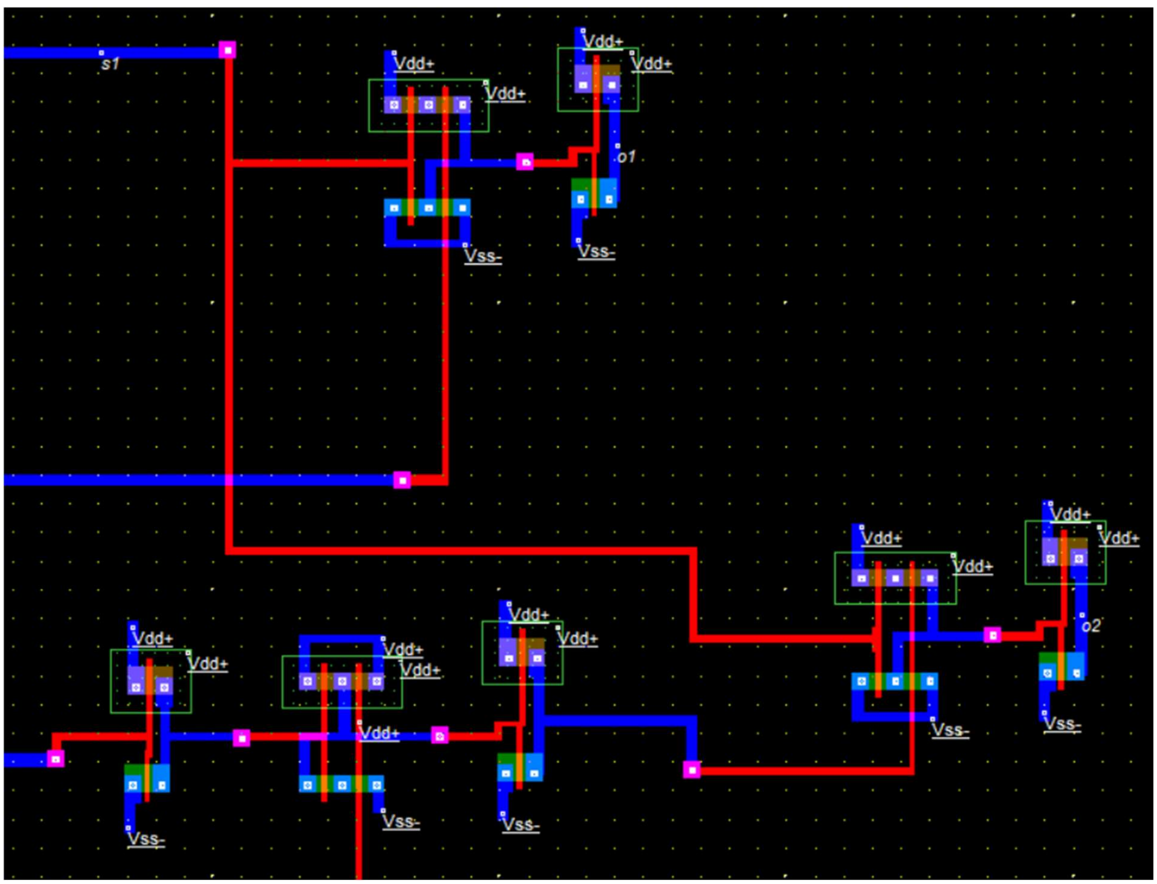

Fig. 7: Layout of Priority Encoder

3.2 Comparator Layout

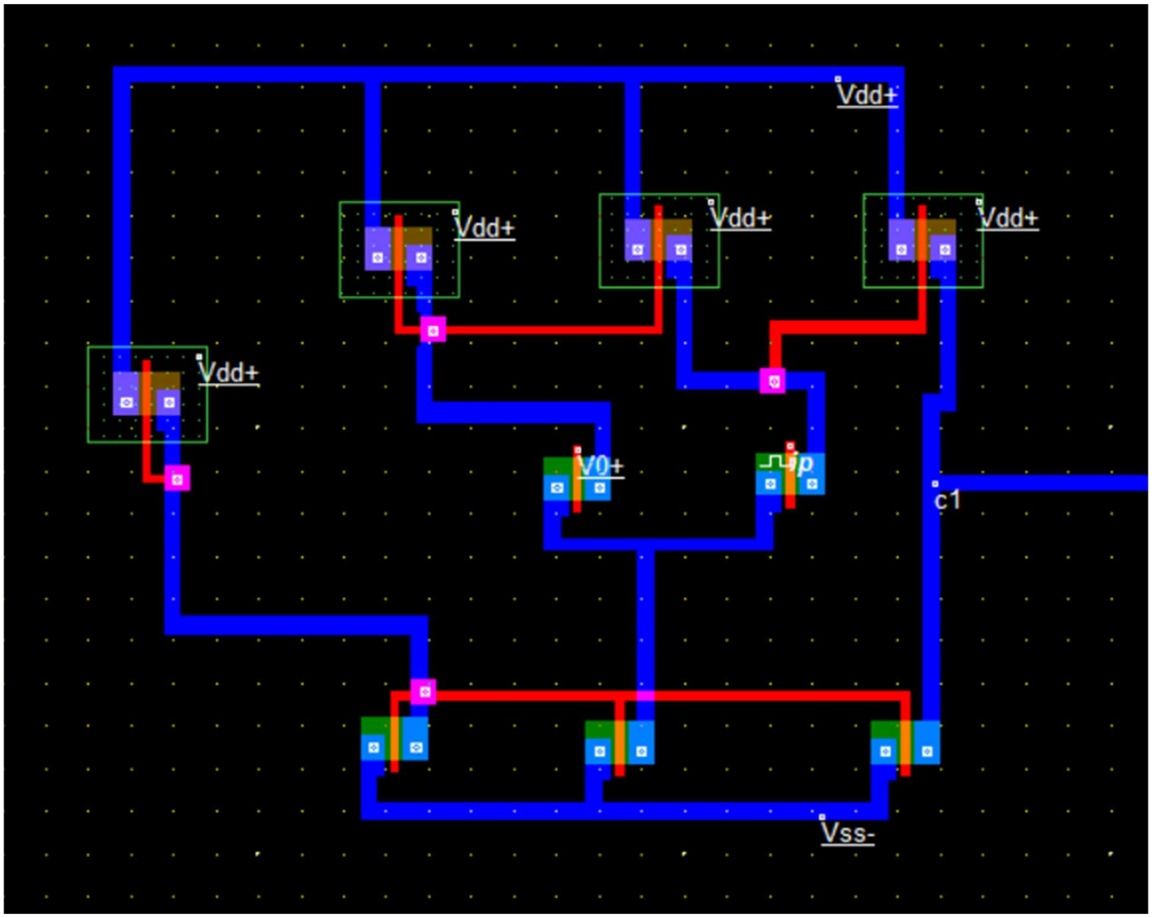

Fig. 8: Layout of Comparator

3.3 Complete ADC Layout

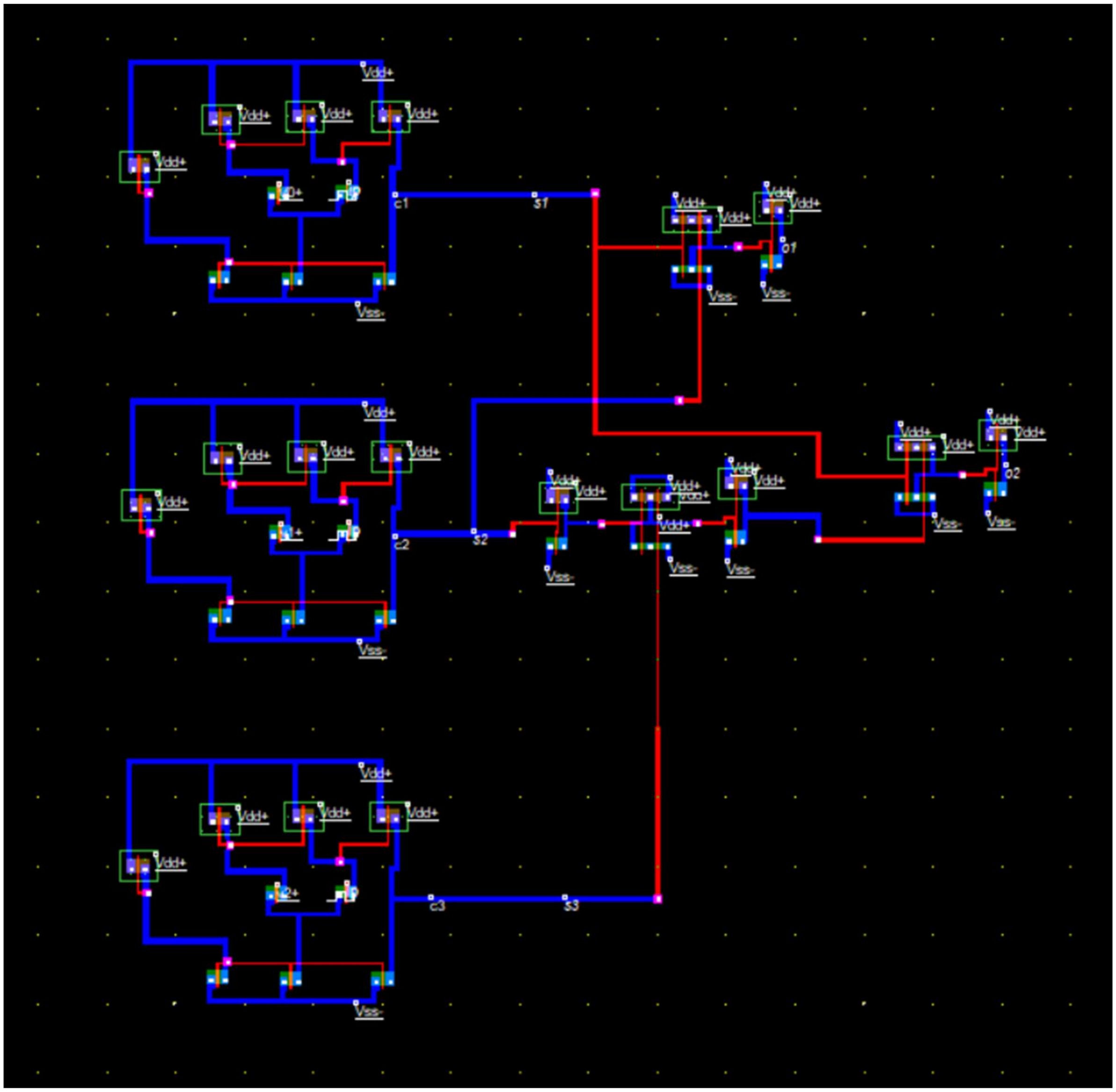

Fig. 9: Complete ADC Layout

3.4 Simulation Results

- Input: Analog ramp / triangular wave
- Output: Digital binary

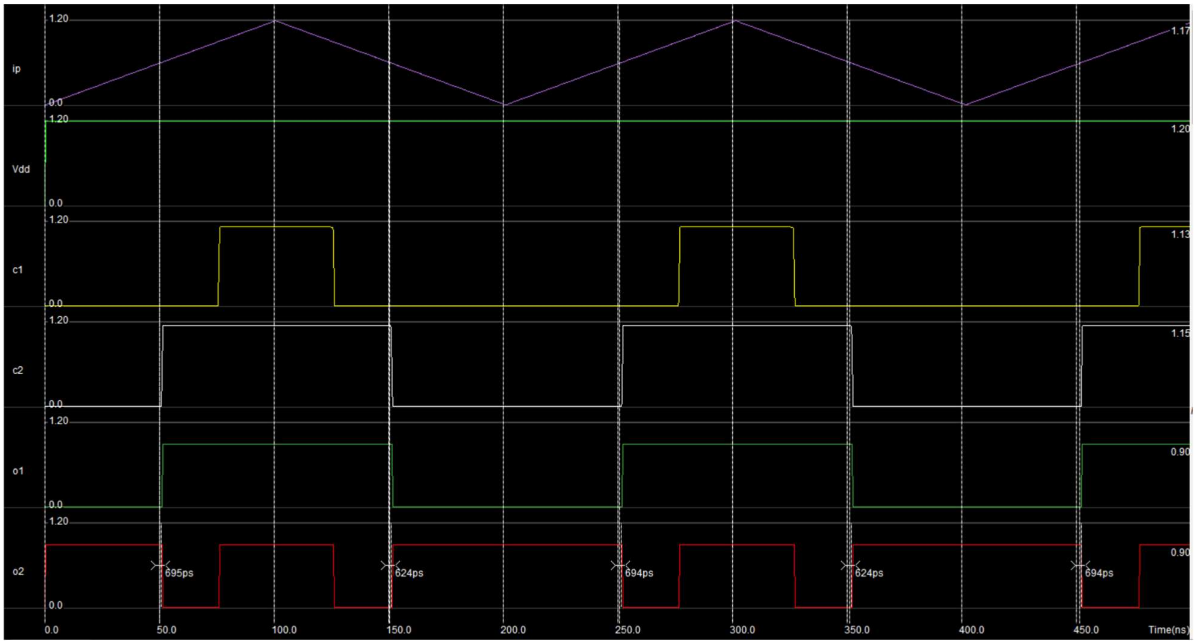

Fig. 10: Simulation Results

3.5 Observations

- Fast conversion achieved
- Correct binary output for all input ranges
- No major delay observed due to small circuit size

Chapter 4: Future scope and applications

Future Scope

1. Can be extended to higher-bit ADC (4-bit, 8-bit)
2. Optimization of power consumption
3. Improved comparator design for accuracy

Applications

1. High-speed data acquisition systems
2. Digital oscilloscopes
3. Communication systems
4. Signal processing units

Chapter 5: References

- Microwind & DSCH Tool Documentation
- Lecture Notes (ECL 312 CMOS Design)
- IEEE Papers on Flash ADC
- Debjyoti-Banerjee/Low-Power-Comparator-using-CMOS-technology
- https://youtu.be/kkZhaDw3DUM?si=hLN1g8AiXWCeiFW0